

Rocket Elevators

ELEVATE SAFETY, SERVICE AND VALUE

INTERNAL DOCUMENT

Hydraulic Elevator Maintenance Procedures

Internal document. Last updated April 2026. Contact: Field Operations Manager, Rocket Elevators Ontario Division.

This document covers the maintenance procedures, schedules, and technical guidance for hydraulic elevators in our fleet. Hydraulic systems make up a significant portion of our managed devices, primarily in low-rise buildings across Ontario. They include Passenger Elevators in buildings of 2 to 6 stories, Freight Elevator-P (hydraulic/pneumatic) variants, LULA elevators, and Sidewalk Elevators.

KONE is our primary contractor for hydraulic systems and modernizations. Our 14 field technicians perform routine checks and minor maintenance, but anything affecting safety circuits, structural integrity, or rated capacity must be handled by KONE or another licensed contractor. This document defines the boundary between what our technicians handle in the field and what gets escalated.

All hydraulic elevator maintenance must comply with Ontario Regulation 209/01 and the CSA B44 Safety Code for Elevators and Escalators, as enforced by the Technical Standards and Safety Authority (TSSA). For machine room and pit requirements specific to CSA B44, see the Safety Code Quick Reference.

How Hydraulic Elevators Work

A hydraulic elevator moves the car by pushing hydraulic fluid under pressure. The main components are:

- **Pump unit:** a motor and pump assembly in the machine room that pressurizes the hydraulic fluid
- **Control valve:** regulates fluid flow to control car speed, direction, and stopping. The valve opens to allow fluid into the cylinder (car goes up) and opens a return path to let fluid drain back to the reservoir (car goes down). Gravity, not the pump, brings the car down
- **Jack (piston and cylinder):** the hydraulic cylinder that converts fluid pressure into mechanical motion. The piston extends to raise the car and retracts as fluid returns to the reservoir
- **Reservoir (tank):** holds the hydraulic fluid supply. Fluid level changes as the car moves; it drops when the car is at the top floor and rises when the car is at the bottom
- **Piping and fittings:** connect the pump unit, control valve, and jack. Leaks at fittings are one of the most common maintenance issues

Holed vs. Holeless Configurations

Our fleet includes two hydraulic configurations, and the distinction matters for maintenance:

Holed (in-ground) hydraulic elevators have the cylinder buried in a borehole beneath the elevator pit. The piston extends upward from underground to push the car. These are the older, more common design in our fleet. The underground cylinder introduces corrosion risk, environmental contamination potential, and difficulty inspecting without excavation. See the Underground Cylinder Monitoring section for our protocols.

Holeless hydraulic elevators mount the cylinder(s) above ground in the hoistway. No borehole is required. These are easier to inspect and maintain, and are standard in newer installations. Holeless configurations are the preferred option when building owners undertake modernization.

Some units use a **roped hydraulic** design, where the jack connects to the car through ropes and sheaves rather than pushing directly. This multiplies travel distance, allowing hydraulic systems to serve slightly taller buildings. Roped hydraulic elevators require additional maintenance on ropes and sheaves.

Hydraulic elevators have different failure modes than traction systems. The fluid is the lifeblood: contaminated, degraded, or low fluid causes most problems. Seals wear and leak. Underground cylinders corrode invisibly. Ontario temperature extremes affect fluid viscosity. Every section of this document traces back to these characteristics.

Routine Maintenance Schedule

Our technicians follow a tiered maintenance schedule for every hydraulic elevator. This does not replace TSSA-mandated inspections (see Inspection Types and What They Check), but ensures we catch problems early between inspections. All visits must be logged (see Record-Keeping).

Monthly Checks

Every hydraulic elevator gets a monthly visit from the assigned zone technician (approximately 30 to 45 minutes per device):

1. **Fluid level check:** with the car at the lowest landing, check the reservoir sight glass or dipstick. Record the level and compare to last month. A gradual decline indicates a leak. Note whether the reading is hot or cold (see Hydraulic Fluid Management).
2. **Control valve operation:** ride the car through a full cycle (bottom to top, top to bottom). Listen for unusual sounds. Check that the car starts, accelerates, decelerates, and stops smoothly.
3. **Ride quality assessment:** during the ride cycle, check for vibration, unusual noise, and leveling accuracy at each floor.
4. **Door operation:** test doors at every landing. Check speed, nudging operation, and door restrictors. Verify the car does not move with doors open. Door timing adjustments are a minor repair our technicians can handle.
5. **Pit condition:** inspect the pit for standing water, oil accumulation, debris, and structural deterioration. Check the pit stop switch and pit light. Oil in the pit may indicate a cylinder seal leak, especially on in-ground units.
6. **Emergency communication:** test the emergency phone or intercom. Confirm it connects to the monitoring service.
7. **Car operating panel:** check all buttons, indicators, and displays. Replace burned-out indicators or non-responsive buttons (minor repairs our technicians handle directly).
8. **Machine room condition:** check cleanliness, ventilation, temperature, and that no unauthorized items are stored in the room.

Quarterly Checks

Performed in addition to monthly checks (adds 30 to 60 minutes):

1. **Pressure test:** perform static and running pressure tests. Record readings and compare to rated operating pressure. See Pressure System Procedures for detailed procedures.
2. **Fluid temperature under load:** after 5 to 10 consecutive round trips, measure fluid temperature at the reservoir. Normal range is 30 to 60 degrees Celsius. Above 65 degrees consistently requires investigation (see Temperature Considerations).
3. **Pump motor current draw:** measure motor current during a loaded up-run. Compare to nameplate rating. Current significantly above nameplate may indicate pump issues, a blocked filter, or degraded fluid.
4. **Safety device testing:** test the car safety, oil/spring buffer in the pit, and slack rope device (roped hydraulic units). Verify emergency stop switches. Safety device testing involving safety circuits is contractor work; our technicians verify devices are present and functional, then escalate if needed.

Semi-Annual Checks

Performed twice per year, typically spring and fall:

1. **Comprehensive fluid analysis:** draw a fluid sample and send to the lab for viscosity, water content, acidity, particulate contamination, and additive depletion testing. Results in 5 to 7 business days. See Hydraulic Fluid Management for interpretation.
2. **Cylinder and jack seal inspection:** inspect the piston rod where it exits the cylinder head for oil weeping or leaking. On holeless units, inspect the full cylinder. On in-ground units, inspect the visible portion above the pit floor. Seal weeping beyond a light film requires escalation to KONE.
3. **Control valve calibration check:** verify leveling accuracy within 6 mm (1/4 inch) at every floor. Check speeds against rated speed. Deviations require escalation to KONE. Our technicians do not adjust control valves.
4. **Piping and fitting inspection:** inspect all visible hydraulic lines and connections for leaks, corrosion, or damage. Tighten minor accessible fittings only. High-pressure fittings are contractor work.

Annual Checks

Comprehensive system review, often coinciding with TSSA periodic inspections:

1. **Full system performance test:** run the car under no-load and full-load conditions. Measure travel time, leveling accuracy at every floor, door times, and ride quality. Compare to design specifications and previous annual results.
2. **Rope and sheave inspection (roped hydraulic only):** inspect ropes for broken wires, wear, corrosion, and tension. Inspect sheaves for groove wear and bearing condition. Replacement is contractor work; our technicians document the condition.
3. **Pit equipment inspection:** inspect buffer, pit stop switch, pit light, pit ladder, sump pump, and governor tension sheave where applicable. Check for water intrusion or structural issues.
4. **Machine room equipment review:** inspect pump unit, motor, starter/contacter, control valve, reservoir, piping, controller, and disconnect switch for leaks, wear, corrosion, or overheating.
5. **Documentation review:** verify the data plate, wiring diagrams, maintenance records, and TSSA registration are current and legible.

Hydraulic Fluid Management

Most hydraulic elevator problems trace back to fluid that is low, contaminated, degraded, or the wrong type.

Fluid Types

Our fleet uses two categories:

Petroleum-based hydraulic oil is the standard in most of our fleet. It provides excellent lubrication and has well-understood performance characteristics. Most existing systems were designed for petroleum-based fluid.

Biodegradable hydraulic fluid (vegetable-based or synthetic ester) is used in some newer installations where environmental sensitivity is a concern (buildings near waterways, facilities with in-ground cylinders). It costs more and has a shorter service life, but reduces environmental liability if a leak occurs.

Never mix petroleum-based and biodegradable fluids. Mixing causes seal degradation, sludge formation, and loss of lubrication. If switching fluid types, KONE must perform a full system flush before refilling.

Checking Fluid Levels

Fluid level depends on temperature and car position, so consistency matters:

- **Standard reading position:** car at the lowest landing, system idle for at least 15 minutes
- **Cold reading** (below 20 degrees Celsius): fluid occupies less volume, so the level reads lower
- **Hot reading** (above 40 degrees Celsius): fluid expands, so the level reads higher
- Always compare cold to cold and hot to hot across months

Typical capacities: residential passenger elevator holds 150 to 300 litres; freight and higher-capacity units hold 400 to 800 litres. Exact capacity is on the data plate in the machine room.

Fluid Testing and Analysis

Quarterly checks are visual: fluid color through the sight glass (clear amber is healthy; dark brown, cloudy, or milky indicates problems) and smell at the reservoir cap (burnt smell indicates overheating).

Semi-annual lab analysis tests for:

- **Viscosity:** fluid that has thinned loses sealing and lubrication; fluid that has thickened increases pump effort and slows valve response
- **Water content:** causes corrosion and reduces lubrication. Above 0.1% is a concern; above 0.5% requires immediate attention
- **Acidity (Total Acid Number):** measures oxidation and degradation. A TAN exceeding the manufacturer's threshold means additives are depleted
- **Particulate contamination:** accelerates wear on seals, valves, and pump components. Lab reports ISO 4406 cleanliness class; our threshold is 18/16/13
- **Additive depletion:** when anti-wear and anti-oxidation additive levels drop below 50% of new-fluid values, the fluid is approaching end of life

When to Replace Fluid

Replace based on condition, not calendar:

- Lab results show viscosity, acidity, or contamination outside acceptable ranges
- Water content exceeds 0.5% and cannot be resolved by fixing the intrusion source
- Fluid has been in service more than 7 to 10 years without a full change
- Major contamination event (water intrusion, seal failure, incorrect fluid added)

Fluid replacement is contractor work. KONE drains, disposes of old fluid, cleans the reservoir, and refills.

Disposal Requirements

Used hydraulic fluid is regulated waste under Ontario's Environmental Protection Act. It must not be poured down drains, disposed of with regular waste, or left on site. Contractors handle disposal through licensed waste haulers with documented chain of custody. If our technicians encounter spilled or improperly stored fluid, report it to the Field Operations Manager immediately.

Pressure System Procedures

Our technicians perform pressure tests quarterly, and any time a hydraulic elevator exhibits performance issues (slow travel, drift, noise), a pressure test should be the first diagnostic step.

Equipment

Tests require a calibrated hydraulic pressure gauge (our standard kit: 0 to 5000 psi / 0 to 350 bar). The gauge connects to the test port on the pump unit or control valve manifold. If no test port exists, escalate to KONE for installation.

Static Pressure Test

The static pressure test measures system pressure with the car stationary under load:

1. Position the car at the top floor with a representative load (at least 50% of rated capacity if possible; otherwise, note the load condition).
2. Connect the pressure gauge to the test port.
3. Allow the system to stabilize for at least 2 minutes after the car has stopped.
4. Record the static pressure reading.
5. Monitor the pressure for 5 minutes. A stable reading indicates good seal integrity. A gradual drop indicates an internal leak, either at the cylinder seals or the control valve.

Normal static pressure depends on the system design and rated load, but most passenger elevators in our fleet operate in the 200 to 500 psi range (14 to 35 bar). The system's design pressure is on the data plate.

Running Pressure Test

The running pressure test measures system pressure while the car is traveling upward under load:

1. With the gauge still connected, send the car on a full up-run under load.
2. Record the peak pressure during the run.

3. Compare to the rated working pressure on the data plate.

Running pressure should be within 10% of the rated working pressure. Pressure significantly below rated may indicate a worn pump, internal valve bypass, or low fluid. Pressure significantly above rated may indicate a restriction in the hydraulic line or an improperly adjusted relief valve.

Relief Valve Test

The relief valve is a critical safety device that prevents dangerous overpressure. It should open (bypass fluid back to the reservoir) at a pressure set above the normal working pressure but below the system's maximum rated pressure. Relief valve settings are specified on the data plate.

1. With the car at the lowest landing, place a load in the car exceeding the rated capacity by approximately 25% (coordinate with the building owner for test weights).
2. Initiate an up-run. The relief valve should prevent the car from lifting the excessive load by bypassing fluid.
3. Record the pressure at which the relief valve activates.
4. Compare to the specified relief valve setting.

If the relief valve activates at a pressure significantly different from the specification (more than 5% deviation), escalate to KONE for adjustment. Our technicians do not adjust relief valves. A relief valve that fails to activate is a critical safety issue; tag the device out of service and contact KONE immediately. See the operations handbook for the TSSA shutdown response procedure.

What Abnormal Readings Indicate

READING	POSSIBLE CAUSE	ACTION
Static pressure drops over time	Internal seal leak at cylinder or valve bypass	Escalate to KONE for seal inspection
Running pressure below rated	Worn pump, low fluid, air in system	Check fluid level first; if normal, escalate
Running pressure above rated	Restricted line, failing relief valve	Escalate to KONE immediately
Relief valve activates at wrong pressure	Incorrect valve setting, spring fatigue	Escalate to KONE for adjustment
Pressure fluctuates erratically	Air in system, loose gauge connection, failing pump	Verify gauge connection; if solid, check fluid level and escalate

Jack and Cylinder Maintenance

Jack maintenance focuses on seal integrity, cylinder condition, and, for in-ground units, corrosion and environmental risk.

Seal Inspection

Seals prevent fluid from leaking past the piston. Over time, they wear through a predictable progression:

1. **Normal:** no visible fluid on the piston rod above the seal.
2. **Light weeping:** a thin film of oil on the piston rod. This is normal in small amounts and actually helps lubricate the seal. Monitor and note in the maintenance log.
3. **Active weeping:** visible oil drips or accumulation around the cylinder head. The car may begin to exhibit slow drift (settling below floor level when stopped). Escalate to KONE for assessment.
4. **Active leak:** fluid pooling in the pit or on the machine room floor. The car drifts noticeably. Fluid level in the reservoir drops measurably between visits. This is an urgent maintenance issue. Escalate to KONE for jack repacking or cylinder replacement.

During every pit inspection, check the pit floor for fresh oil. On in-ground cylinders, oil in the pit may be coming from the underground portion of the cylinder, which is more serious (see Underground Cylinder Monitoring).

Jack Repacking

When seals reach the active weeping or leaking stage, the jack needs to be **repacked** (seals replaced). This is always contractor work performed by KONE. The process involves taking the elevator out of service, removing the cylinder head, extracting old seals, inspecting the piston rod for scoring, installing new seals, and testing.

If the piston rod is scored, simple repacking will not solve the problem; the rod may need to be polished, re-chromed, or replaced. Our technicians should note any visible scoring during routine inspections, as catching it early avoids more expensive repairs.

Underground Cylinder Concerns

In-ground cylinders are buried and cannot be visually inspected. The primary concerns are corrosion (from soil moisture, groundwater, and road salt), cylinder type (**single-bottom** vs. **double-bottom**), and environmental contamination risk. Double-bottom cylinders (also called double-cased) have an outer protective casing with a monitoring space that can be tested for fluid, indicating inner cylinder failure before soil contamination. Older installations often have single-bottom cylinders; if records do not specify, assume single-bottom.

A failed underground cylinder can release fluid into soil and groundwater, creating substantial environmental liability. See Underground Cylinder Monitoring for our detailed protocols.

Pump Unit and Motor Maintenance

The pump unit runs every time the car travels upward and sits idle during downward travel (gravity returns the car). Frequent start/stop cycling causes wear on the motor, pump, and electrical components.

Motor Temperature

The motor's nameplate specifies its insulation class and maximum operating temperature (Class B: 130 degrees Celsius; Class F: 155 degrees Celsius). Use a non-contact infrared thermometer during quarterly

visits. Surface temperature is typically 10 to 20 degrees below internal winding temperature. If surface temperature exceeds 80 degrees Celsius after a normal duty cycle, investigate.

Persistent overheating can indicate: low fluid level, degraded fluid (increased viscosity), electrical supply issues (low voltage, phase imbalance), worn pump internals, or excessive duty cycle.

Bearing Inspection

Early signs of bearing wear include grinding/squealing/rumbling sounds, excessive vibration, and bearing housings significantly hotter than the motor body. Note concerns during routine visits; bearing replacement is contractor work.

Pump Noise Diagnosis

Changes in pump noise are diagnostic:

- **Whining or high-pitched noise:** often indicates cavitation (pump starving for fluid). Check fluid level first; if adequate, the suction strainer may be clogged.
- **Knocking or hammering:** may indicate air in the system, worn pump internals, or loose mounting. Check for air bubbles in the sight glass and pump mounting bolts.
- **Increased overall noise level:** gradual increase suggests wear; sudden increase warrants immediate investigation.

Starter and Contactor Inspection

Visually inspect contactors during quarterly visits:

- Look for pitting, burning, or erosion on contactor tips
- Check for loose connections at terminal blocks
- Listen for chattering (rapid on-off cycling), which indicates a control circuit problem
- Check that the overload relay matches the motor current rating

Contactors replacement and electrical work beyond visual inspection is contractor work. Escalate to KONE if visible damage or overload tripping occurs.

Control Valve Maintenance

The control valve determines car speed, start/stop smoothness, and leveling accuracy. Valve performance directly affects ride quality, which building occupants notice immediately.

Leveling Accuracy

Leveling is how precisely the car floor aligns with the landing floor when stopped. CSA B44 requires accuracy within 12 mm (1/2 inch) for passenger elevators. Our internal standard is tighter: within 6 mm (1/4 inch).

Consistent mis-leveling may indicate valve drift, worn valve components, temperature-related fluid changes, or mechanical binding in guide rails. Document which floor, how far off, and which direction; escalate to KONE if deviation exceeds 6 mm.

Speed Adjustment

Up-speed is governed by pump flow through the valve; down-speed by how fast the valve allows fluid to return to the reservoir. Rated speeds for hydraulic passenger elevators range from 0.25 to 0.75 metres per second (50 to 150 feet per minute). Freight elevators are generally 0.12 to 0.50 metres per second.

Measure travel time during annual performance tests. Speed changes exceeding 10% from rated or from the previous year warrant escalation for valve adjustment.

Valve Response and Ride Quality

If ride quality complaints increase, ride the car and characterize the issue:

- **Hard start:** car lurches when beginning to move. May indicate a sticky valve or abrupt opening profile.
- **Rough stop:** car bumps or overshoots. May indicate valve closing too quickly or hydraulic hammer.
- **Slow leveling:** car takes several seconds of small adjustments to reach final position. May indicate worn valve or incorrect re-leveling settings.

All valve adjustments are contractor work. Document specifics (direction, floors, what the passenger feels) so KONE can diagnose efficiently.

Re-leveling

Hydraulic elevators can experience **drift**, where the car slowly settles below floor level after stopping due to fluid leaking past internal seals. Modern systems have automatic **re-leveling**: when drift exceeds a threshold (typically 12 to 25 mm), the pump activates briefly to correct.

Frequent re-leveling (audible as brief pump runs while the car is stationary) indicates increasing internal leakage. Not an emergency, but document the frequency so we can plan repair before drift becomes noticeable to passengers.

Temperature Considerations

Ontario's climate creates challenges for hydraulic systems. Machine rooms can drop below freezing in winter and exceed 40 degrees Celsius in summer. Fluid performance is directly tied to temperature.

Cold Weather (Below 10 Degrees Celsius)

Cold fluid has increased viscosity (thicker), causing slower travel speed, increased pump effort (potentially tripping motor overload), and sluggish valve response affecting leveling accuracy.

Many machine rooms have **immersion heaters** in the reservoir to maintain fluid above 10 to 15 degrees Celsius. During fall visits, verify the heater is functional:

1. Check thermostat setting (minimum 15 degrees Celsius).
2. Verify the heater element works by checking fluid temperature after a cold night.
3. Inspect heater wiring for damage or corrosion.
4. If the machine room lacks a heater and drops below 5 degrees Celsius in winter, recommend installation to the building owner.

Hot Weather (Above 30 Degrees Celsius)

Hot fluid thins (viscosity decreases), causing increased internal leakage and drift, reduced pump efficiency, and accelerated fluid degradation through oxidation.

Many machine rooms have **ventilation fans** to manage heat. During spring visits:

1. Verify fans are operational on the correct thermostat setting.
2. Check that air intake vents are not blocked.
3. If the machine room consistently exceeds 35 degrees Celsius in summer, recommend improved ventilation.

Temperature Monitoring

Record machine room ambient temperature and fluid temperature during every visit. Over time, this data reveals seasonal patterns and identifies machine rooms needing heating or cooling improvements.

Common Hydraulic Issues Quick Reference

Quick diagnostic reference for the most common hydraulic issues.

SYMPTOM	LIKELY CAUSE	FIELD TECH ACTION	ESCALATE TO KONE?
Car drifts down when stopped at floor	Cylinder seal wear or control valve bypass	Check if re-leveling is active; measure drift rate over 5 minutes; record findings	Yes, if drift exceeds 12 mm in 10 minutes
Slow up-travel speed	Low fluid level, cold fluid, worn pump, partially closed valve	Check fluid level and temperature; compare to rated speed	Yes, if fluid level and temperature are normal
Noisy pump (whining)	Cavitation from low fluid or clogged suction strainer	Check fluid level; check for air bubbles in sight glass	Yes, if fluid level is adequate
Noisy pump (knocking)	Air in system, worn pump, loose mounting	Check pump mounting bolts; check for air in sight glass	Yes, regardless of bolt condition
Car won't level accurately	Control valve mis-calibration, worn valve, temperature fluctuation	Measure leveling deviation at each floor; record fluid temperature	Yes, if deviation exceeds 6 mm
Oil smell or visible oil in pit	Cylinder seal leak, fitting leak, packing leak	Identify leak source if visible; check fluid level; photograph	Yes, always
Car runs rough (jerky ride)	Valve adjustment, air in system, guide rail issue	Ride the car and document the specific behavior (which direction, which floors)	Yes, with documented observations
Motor overheating	Low fluid, degraded fluid, electrical issue, high duty cycle	Check fluid level and condition; check motor current draw	Yes, if fluid is not the cause
Car does not respond to call	Electrical control issue, pump failure, safety circuit tripped	Check for tripped breakers or safety circuits; reset if safe to do so	Yes, if reset does not resolve
Relief valve leaking (audible hissing at pump unit)	Relief valve not seating properly, incorrect setting	Do not attempt adjustment; document the sound and conditions	Yes, immediately

For hydraulic-specific failure patterns and statistical trends across our fleet, see Common Failure Modes by Elevator Type.

Underground Cylinder Monitoring

Underground cylinders are one of our most significant maintenance and liability concerns. The cylinder is buried beneath the pit, typically in a steel casing (or, in older installations, directly in soil contact). We cannot see the cylinder without excavation, so we rely on indirect indicators.

A corroded cylinder can fail two ways: **sudden failure (blowout)**, where the wall ruptures causing uncontrolled car descent (see Emergency Response Protocols for blowout response), or **gradual leak**, where

small perforations allow fluid to seep into soil undetected for months. Cleanup costs for soil remediation can exceed \$100,000. Both scenarios create liability for Rocket Elevators.

Risk Factors

Not all in-ground cylinders carry equal risk. The following factors increase the likelihood of corrosion and failure:

- **Age:** cylinders installed before 1990 are at higher risk. Cylinders more than 30 years old warrant closer monitoring regardless of other factors.
- **Single-bottom construction:** no protective outer casing. Fluid-to-soil contact if the cylinder fails.
- **High water table:** groundwater in contact with the cylinder accelerates corrosion.
- **Corrosive soil conditions:** areas with high salt content (proximity to major roads where road salt accumulates), acidic soils, or industrial contamination.
- **No cathodic protection:** older installations often lack corrosion-prevention systems.
- **History of fluid loss:** unexplained drops in reservoir level that do not correspond to visible leaks.

Monitoring Indicators

Our technicians should watch for these indicators during every visit to a building with in-ground hydraulic cylinders:

1. **Unexplained fluid loss:** reservoir level drops between visits with no visible leak in the machine room, pit, or hoistway. This is the primary indicator of an underground cylinder leak. Track fluid level readings month over month. A consistent downward trend that cannot be attributed to visible leaks or normal system operation is a red flag.
2. **Ground settlement:** sinking, cracking, or soft spots in the ground around the elevator shaft or pit area. Fluid leaking into soil can dissolve or displace material, causing the ground to settle.
3. **Oil odor:** a petroleum smell in or around the pit area that does not correspond to a visible leak source.
4. **Fluid contamination:** lab analysis showing unexpected contaminants (soil particles, water with mineral content inconsistent with condensation). Contaminated fluid can indicate that the cylinder wall has breached in both directions.
5. **Pump run time increase:** if the pump is running more frequently or for longer durations to maintain pressure, fluid may be leaking from the system.

Response Protocol

If monitoring indicators suggest a possible underground cylinder issue:

1. Document all findings with photographs, measurements, and dates.
2. Report to the Field Operations Manager immediately.
3. Increase monitoring frequency to bi-weekly for the affected device.
4. The Field Operations Manager engages KONE for a formal cylinder assessment, which may include ultrasonic testing of accessible portions, pressure decay testing, or ground-penetrating analysis.
5. Provide the building owner with a written risk assessment and recommendation.

Holeless Conversion

For buildings with aging in-ground cylinders, we routinely recommend that building owners consider **holeless conversion**: replacing the underground cylinder with an above-ground cylinder configuration. This eliminates the corrosion and contamination risk entirely.

Holeless conversion is a major alteration requiring TSSA approval. It involves removing the existing jack and underground cylinder (including environmental assessment), installing above-ground cylinders, modifying the car frame and guide rails, and updating the control system. The device status during conversion is "Undergoing Major Alt" in our system. KONE handles the project; our role is to advise building owners and coordinate scheduling with KONE and TSSA.

Record-Keeping

Maintenance records track device condition over time, prepare us for TSSA inspections, and feed into risk prediction models. Every visit must be documented.

What to Record

After every visit, log:

1. **Device identification**: device ID, building address, device type, configuration (holed/holeless, roped/direct)
2. **Visit type**: monthly, quarterly, semi-annual, annual, or off-schedule callback
3. **Measurements**: fluid level (noting hot or cold), fluid temperature, machine room temperature, pressure readings, motor current draw, leveling accuracy, travel time (as applicable to visit type)
4. **Observations**: ride quality, noise, vibration, visual condition of components, pit and machine room condition
5. **Actions taken**: minor repairs performed, adjustments made, fluid added
6. **Issues identified**: problems found, severity (informational, needs monitoring, needs scheduled repair, urgent), whether escalated to KONE
7. **Fluid sample**: sample date and lab reference number if taken
8. **Photos**: oil leaks, damaged components, pit conditions, anything useful to KONE or TSSA

Where Records Go

All records enter the company database, which tracks visit history, trend data for quantitative measurements, open issues, contractor work orders, and lab results per device.

How Records Inform Operations

- **Inspection preparation**: review maintenance history before TSSA inspections. Address open issues before the inspector arrives. Complete records demonstrate proper maintenance.
- **Risk prediction**: patterns like increasing re-leveling frequency, declining fluid condition, or rising machine room temperatures feed into risk models across our 40,000 devices.
- **Lifecycle planning**: maintenance history supports repair vs. modernization vs. replacement decisions for building owners.

- **Incident context:** if an incident occurs, records show whether there were warning signs and whether the maintenance schedule was followed.

Technicians play a direct role in data quality. "Checked fluid, ok" is far less useful than "fluid level at 72% on cold reading (18C), consistent with January reading of 74%. Clear amber color, no odor." Specific records enable trend analysis and risk prediction.